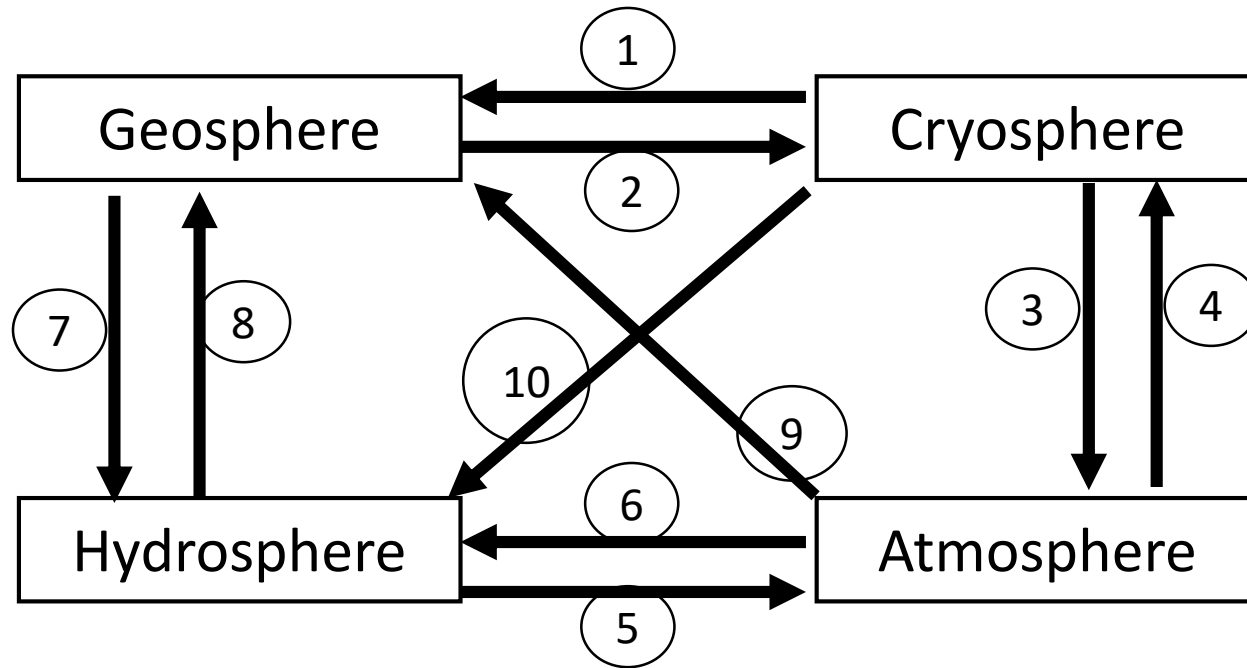




INTERACTION OF SPHERES



Recap



1. Glacial erosion
2. Volcanic/Tectonic activity/ Permafrost
3. Influence Temperature/Albedo effect
4. Wind erosion/Snow/ High temperature causes melting & runoff
5. Evaporation/Water cycle/ Gas exchange CO₂
6. Precipitation/Heat exchange/ /ocean currents
7. Topography – path for rivers and streams/ Mineral flux
8. Groundwater/ Caves/River erosion
9. Wind erosion/Acid rain
10. Rise in water level/Thermohaline circulation

Geosphere ↔ Atmosphere

Geosphere to Atmosphere:

- **Volcanic Outgassing:** Volcanic eruptions discharge carbon dioxide, sulfur dioxide, water vapor, and ash into the atmosphere. Major explosive eruptions push aerosols into the stratosphere, blocking sunlight and temporarily cooling global temperatures.
- **Dust Transport:** Wind erosion picks up fine soil particles, mineral dust, and sand from deserts (e.g., the Sahara) and carries them thousands of miles through the atmosphere, providing condensation nuclei for clouds.

Atmosphere to Geosphere:

- **Atmospheric Weathering:** Acidic rain (formed when atmospheric CO₂ dissolves in rainwater) chemically weathers silicate and carbonate rocks on the ground.
- **Wind Erosion:** Strong winds physically carve geological structures, shape sand dunes, and erode surface rocks over millions of years.



Geosphere ↔ Hydrosphere

Geosphere to Hydrosphere:

- **Chemical & Mineral Runoff:** Land erosion releases salts, calcium, phosphorus, and iron into rivers, which deliver them to the oceans. This mineral flux regulates ocean chemistry and salinity.
- **Seafloor Topography:** Submarine mountain ranges (mid-ocean ridges) and ocean trenches shape ocean basins, dictating deep-sea currents and circulation patterns.

Hydrosphere to Geosphere:

- **Hydrated Subduction:** As oceanic tectonic plates sink under continental plates, they carry water-saturated sediments deep into Earth's mantle. This water lowers the melting point of mantle rocks, triggering volcanism.
- **Coastal and River Erosion:** Waves, tides, and river currents continually reshape landmasses, carving canyons, deltas, and coastlines.



Geosphere ↔ Cryosphere

Geosphere to Cryosphere:

- **Topographic Elevation:** Tectonic uplift builds high mountain ranges (e.g., the Himalayas, Andes), creating colder, high-altitude climates that allow mountain glaciers and ice caps to form.
- **Geothermal Heat Flux:** Geothermal heat from the Earth's crust warms the bases of glaciers and ice sheets, creating subglacial lakes and enabling ice movement.

Cryosphere to Geosphere:

- **Glacial Gliding and Plucking:** Heavy ice sheets scrape across the bedrock as they move, carving U-shaped valleys, deep fjords, and leaving behind glacial till and moraines. (Glacial erosion)
- **Post-Glacial Rebound (Isostasy):** When massive ice sheets melt, the heavy weight pressing down on the Earth's crust is lifted, causing the geosphere to slowly rise back up over thousands of years.



Hydrosphere ↔ Atmosphere

Hydrosphere to Atmosphere:

- **Evaporation and Latent Heat:** Ocean and surface water evaporates into water vapor (a potent greenhouse gas), storing heat energy that fuels weather systems, storms, and hurricanes.
- **Gas Exchange:** Oceans absorb atmospheric gases like oxygen and carbon dioxide, releasing them back when surface temperatures rise.

Atmosphere to Hydrosphere:

- **Precipitation:** Condensed water vapor falls as rain, sleet, or hail, directly replenishing lakes, rivers, groundwater aquifers, and oceans.
- **Wind-Driven Surface Currents:** Atmospheric wind patterns (like trade winds and westerlies) push ocean surface waters, driving global oceanic currents.



Hydrosphere ↔ Cryosphere

Hydrosphere to Cryosphere:

- **Ice Shelf Formation:** Cold ocean water freezes along polar coastlines to form **sea ice**, and ocean currents support floating ice shelves extending from land ice.
- **Snowfall Supply:** Moisture evaporated from warm oceans provides the water vapor needed for winter snowfall, which builds up land glaciers.

Cryosphere to Hydrosphere:

- **Freshwater Discharge:** Melting icebergs, glaciers, and ice caps release massive volumes of cold freshwater into ocean basins.
- **Thermohaline Circulation:** When ocean water freezes into sea ice, it leaves behind salt (brine rejection). This cold, extremely salty water sinks to the ocean floor, driving the global oceanic "conveyor belt" (thermohaline circulation).



Atmosphere ↔ Cryosphere

Atmosphere to Cryosphere:

- **Precipitation & Glaciation:** Freezing atmospheric temperatures allow water vapor to condense directly into snow, feeding ice sheet mass.
- **Thermal Melting:** Rising global air temperatures transfer sensible heat to ice surfaces, causing surface melting and runoff.

Cryosphere to Atmosphere:

- **Albedo Effect:** Ice and snow have a high surface reflectivity (albedo), reflecting up to 90% of incoming solar radiation back out into space, keeping the atmosphere cool.
- **Polar High-Pressure Systems:** Extremely cold ice sheets chill the air directly above them, creating dense, sinking cold air masses (polar highs) that influence global wind belts.

